

PERMUTATIONS AND COMBINATIONS – FOUNDATIONAL

Directions for question 1: Select the correct alternative from the given choices.

In how many ways can 10 chocolates be distributed among 5 students, if each student can get any number of chocolates?

☐ a) 500

☐ b) 10^5

☐ c) $10!$

☒ d) 5^{10} (Your answer - ✓)

Solution:

Each chocolate can be given away in 5 ways, i.e., to any of the five students. Hence, the number of ways = 5^{10} .
Choice (D)

Directions for questions 2 and 3: Type in your answer in the input box provided below the question.

In how many ways can 10 identical chocolates be distributed among 5 students, if each student gets at least one chocolate?

Solution:

Let a, b, c, d and e be the number of chocolates that each of the five students gets. $a + b + c + d + e = 10$, where a, b, c, d and e are positive integers. Number of ways of distributing = Number of solutions of the above equation = ${}^{(10-1)}C_{(5-1)} = {}^9C_4 = 126$.
Ans : (126)

Directions for questions 2 and 3: Type in your answer in the input box provided below the question.

If ${}^{10}C_4 - {}^9C_4 = {}^nC_3$, find n.

Solution:

$$\begin{aligned} {}^{10}C_4 - {}^9C_4 &= \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} - \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} \\ &= \frac{9 \times 8 \times 7 \times (10-6)}{4 \times 3 \times 2 \times 1} = \frac{9 \times 8 \times 7 \times 4}{4 \times 3 \times 2 \times 1} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} \\ &= {}^9C_3. \end{aligned}$$

This is also equal to nC_3 . Therefore, $n = 9$.

Ans : (9)

Directions for questions 4 to 8: Select the correct alternative from the given choices.

A party was attended by 30 guests. If each guest shook hands with every other guest exactly once, how many handshakes took place at the party?

- ☐ a) 435
- ☐ b) 870
- ☐ c) 60
- ☐ d) 15

Solution:

One handshake happens between two people. Total number of handshakes = Number of ways of selecting two people from 30 people = ${}^{30}C_2 = 435$. Choice (A)

Directions for questions 4 to 8: Select the correct alternative from the given choices.

There are 10 students who registered for a competition. Find the number of ways of selecting at least 2 and at most 9 students to take part in the competition.

- ☐ a) 1024
- ☐ b) 1023
- ☐ c) 1012
- ☐ d) 1010

Solution:

The number of ways of selecting at least 2 and at most 9 students for the competition = ${}^{10}C_2 + {}^{10}C_3 + {}^{10}C_4 + {}^{10}C_5 + {}^{10}C_6 + {}^{10}C_7 + {}^{10}C_8 + {}^{10}C_9 = ({}^{10}C_0 + {}^{10}C_1 + \dots + {}^{10}C_{10}) - ({}^{10}C_0 + {}^{10}C_1 + {}^{10}C_{10}) = 2^{10} - (1 + 10 + 1) = 1012$. Choice (C)

Directions for questions 4 to 8: Select the correct alternative from the given choices.

The number of words (not necessarily meaningful) that can be formed using all the letters of the word PARROT is _____.

- ☐ a) 120
- ☐ b) 240
- ☐ c) 360
- ☐ d) 720

Solution:

PARROT has a total of 6 letters, including 2 R's. The number of words that can be formed = $\frac{6!}{2!} = 360$. Choice (C)

Directions for questions 4 to 8: Select the correct alternative from the given choices.

Find the number of ways of arranging 5 black balls and 3 red balls in a straight line such that no two red balls are adjacent to each other.

- ☐ a) 2880
- ☐ b) 1200
- ☐ c) 12000
- ☐ d) 14400

Solution:

Let us arrange the 5 black balls first.

_ B _ B _ B _ B _ B _

There are 6 gaps (4 in between and 2 at the extreme ends as shown) where we can arrange the red balls. The 3 red balls can be placed in any three of the 6 gaps. The black balls can be arranged in $5!$ i.e., 120 ways.

The 3 red balls can be arranged in 6P_3 , i.e., 120 ways. Hence, the total number of ways is $120 \times 120 = 14400$. Choice (D)

Directions for questions 4 to 8: Select the correct alternative from the given choices.

Find the number of six-digit numbers with unique digits.

- ☐ a) 900000
- ☐ b) 151200
- ☐ c) 136080
- ☐ d) 60480

Solution:

The first digit can be any of the 9 digits from 1 to 9. The second digit can be any of the remaining 8 digits or 0 and hence has 9 possible values. The third digit can be any of the remaining 8 digits. The fourth digit can be any of the remaining 7 digits and so on.
 $\therefore$ The number of six-digit numbers that can be formed = $9(9)(8)(7)(6)(5) = 136080$.
Choice (C)

Directions for question 9: Type in your answer in the input box provided below the question.

In how many ways can a numbered card or a black coloured card be drawn from a pack of cards?

Solution:

In a pack of cards, there are 36 numbered cards out of which 18 are black. Similarly, there are 26 black coloured cards out of which 18 are numbered. Total number of ways of drawing one card = $36C_1 + 26C_1 - 18C_1 = 44$.
Ans : (44)

Directions for question 10: Select the correct alternative from the given choices.

There are 5 chairs in a row numbered 1 to 5 and each chair is to be painted using one of the colours red or blue or green such that no two adjacent chairs have the same colour. Find the number of ways in which the chairs can be painted.

- ☐ a) 32
- ☐ b) 243
- ☐ c) 48
- ☐ d) 64

Solution:

The first chair can be painted in any of the 3 colours. The second chair can be painted in any of the remaining 2 colours. Similarly, the third chair can also be painted in 2 colours and so on.

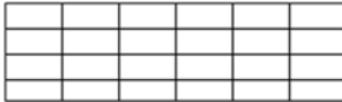
Hence, the total number of ways

$$= 3 \times 2 \times 2 \times 2 \times 2 = 48.$$

Choice (C)

Directions for question 11: Type in your answer in the input box provided below the question.

How many rectangles are there in the 4×6 rectangular grid given below?



Solution:

For a rectangle, we need two vertical lines and two horizontal lines. There are 7 vertical lines and 5 horizontal lines in the given figure.

Selecting 2 lines from 7 vertical lines can be done in 7C_2 ways. Similarly, selecting 2 lines from 5 horizontal lines can be done in 5C_2 ways.

Hence, the total number of rectangles in the figure = ${}^7C_2 \times {}^5C_2 = 210$ Ans : (210)

Directions for questions 12 to 14: Select the correct alternative from the given choices.

Find the number of ways in which 10 letters can be placed in 10 addressed envelopes such that exactly one letter is placed wrongly.

- ☐ a) 133496
- ☐ b) 1334961
- ☐ c) 1
- ☐ d) 0

Solution:

The minimum number of letters that can be wrongly placed is 2. There cannot be a situation where exactly one letter can be placed wrongly. Choice (D)

Directions for questions 12 to 14: Select the correct alternative from the given choices.

Find the number of quadrilaterals that can be formed from 10 points in a plane, out of which exactly 3 points are collinear.

- ☐ a) 220
- ☐ b) 210
- ☐ c) 205
- ☐ d) 203

Solution:

To form a quadrilateral, we need 4 points. Since 3 points are collinear, there cannot be a quadrilateral formed using these 3 points taken together. When the 3 points are taken together, the one other point can be selected in 7 ways. Hence, the number of quadrilaterals formed = ${}^{10}C_4 - 7 = 210 - 7 = 203$.
Choice (D)

Directions for questions 12 to 14: Select the correct alternative from the given choices.

How many six-digit palindromes can be formed using one or more of the digits 1, 2, 3 and 4?

- ☐ a) 4096
- ☐ b) 576
- ☐ c) 64
- ☐ d) 24

Solution:

Let the six-digit palindrome be abccba. Number of six-digit palindromes is the same as the number of ways of choosing the first three digits. Number of ways of choosing the first three digits = $4 \times 4 \times 4 = 64$. Number of six-digit palindromes which can be formed = 64.
Choice (C)

Directions for question 15: Type in your answer in the input box provided below the question.

Find the sum of all the four-digit numbers with distinct digits that can be formed using the digits 2, 3, 4 and 5.

Solution:

$$\text{Sum of all the four-digit numbers} = (2 + 3 + 4 + 5) (1111)3! = 14 (6666) = 93324$$

Ans : (93324)